

How Unified is Your Encoder?

Measuring Representational Convergence in Any-to-Any Vision Models

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Abstract—Any-to-any models such as 4M share encoder weights across modalities, but each modality enters through distinct learned embeddings. We develop a multi-metric protocol (CKA, PWCCA, k-NN overlap) to test whether the encoder converges toward a shared representation, using a mismatched-scene null distribution for significance testing. This characterises the geometric and topological dimensions of unification per modality pair. We complement this with exploratory protocols targeting two blindspots: functional compatibility and ceiling ambiguity.

I. INTRODUCTION

Multimodal models such as 4M [1] rely on a single shared Transformer encoder to process any input modality. While the encoder weights are shared, each modality enters through distinct learned embeddings [1], allowing the network to route information through effectively separate pathways. Zamir [2] has conjectured that such models might exploit this to act as an ensemble of disjoint sub-networks rather than a truly unified encoder.

We frame unification as follows (Fig. 1): each modality projects the scene into a region of the encoder’s latent space. The *intersection* of these projections captures the information common to both modalities (e.g., scene geometry shared between RGB and depth), while their *difference* captures modality-specific information (e.g., color for RGB, absolute scale for depth). We consider the encoder *perfectly unified* if the intersection contains all shared information and the differences are explained entirely by what each modality uniquely provides. In a partially unified encoder, some shared information may remain unaligned across modality subspaces. We acknowledge this is our working definition, motivated by the Platonic Representation Hypothesis [3]; others are possible.

Formally verifying this condition would require quantifying the exact amount of shared information between raw modalities, which remains an open problem in high dimensions [4]. There is therefore no ground truth for the ideal degree of alignment. In practice, no model is likely to be perfectly unified given our definition; the relevant question is *to what degree* unification is achieved. Importantly, unification is not a single axis but a multidimensional spectrum: it may vary across modality pairs (depth-normals may be unified while RGB-segmentation is not), across types of similarity (geometric, topological, functional), and in degree along each dimension.

The core of this project (RQ1) develops a rigorous multi-metric protocol to characterise the geometric and topological dimensions of unification inside 4M’s encoder. This protocol has known blindspots: it cannot assess functional compatibility (whether shared information is encoded interchangeably) nor resolve the ceiling ambiguity (whether unaligned variance is

legitimate or a failure). As a secondary, more exploratory effort (RQ2), we investigate complementary methods that may address these gaps, depending on time constraints.

RQ1. Does the encoder exhibit measurable representational convergence when processing different modalities of the same scene, and is this signal consistent across geometric, directional, and topological similarity metrics?

RQ2. Can complementary protocols (cross-modal transfer, residual analysis) address the functional and ceiling blindspots left open by RQ1?

II. DIAGNOSTIC PROTOCOLS

A. Multi-Metric Convergence Analysis (RQ1)

Methodology. We pass each modality (RGB, depth, normals, segmentation) of a given scene separately into the encoder. This regime is within 4M’s training distribution: the Dirichlet sampling with low α regularly produces mono-modal inputs [1]. For each layer l and pair of modalities (m_i, m_j) , we compute representational similarity over N scenes. Our primary metric is **CKA** [5] (global relational structure, scale-invariant), which operates on $N \times N$ Gram matrices and is therefore robust regardless of activation dimensionality. As complementary checks we use **PWCCA** [6] (directional alignment of principal axes; requires PCA pre-reduction when $N < d$, so results are interpreted with caution) and **k-NN overlap** (local topological similarity; subject to the curse of dimensionality in high- d spaces, mitigated by PCA projection). Together these cover the geometric and topological dimensions of unification. They do *not* measure *functional* compatibility: two spaces may share geometry while encoding information in incompatible formats. Agreement across metrics strengthens conclusions; disagreement reveals which type of similarity is present or absent.

Null Hypothesis Testing. Raw metric values are not directly interpretable: a non-zero CKA between RGB and depth activations could simply reflect architectural biases (shared positional encodings, similar layer norms) or statistical regularities in natural scenes, rather than genuine unification learned by the encoder. To distinguish encoder-driven convergence from these confounds, we test against H_0 : representations share no more similarity than expected by chance. We construct a layer-wise empirical null distribution by computing each metric between modality m_i of scene A and modality m_j of a randomly selected, mismatched scene B , aggregated over enough random pairs for the null distribution to stabilise. This *noise trajectory* captures both architectural biases and baseline dataset similarity, both of which evolve with layer depth. Only

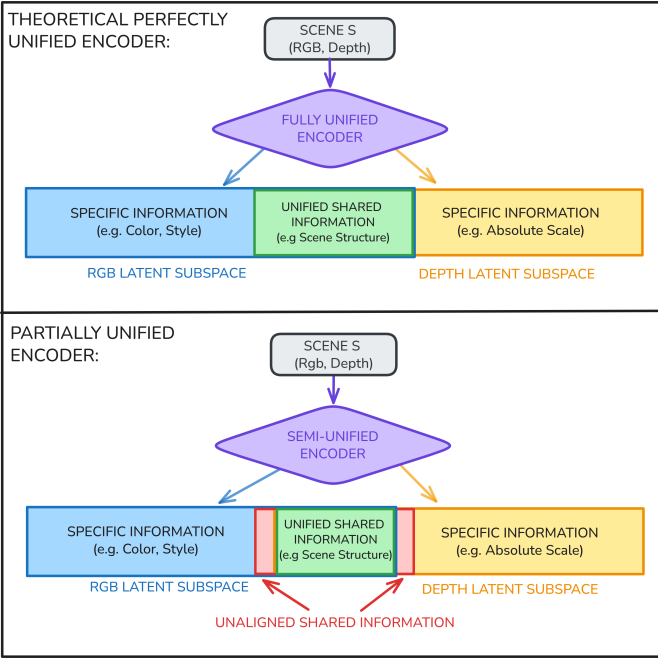


Figure 1. Our working definition of unification. **Top**: a perfectly unified encoder maps all shared information into the intersection of modality subspaces; differences are entirely modality-specific. **Bottom**: a partially unified encoder leaves some shared information unaligned (red regions), encoding it within modality-specific subspaces rather than the shared intersection.

convergence significantly above this floor can be attributed to the encoder learning a shared scene representation.

Outcomes. Fig. 2 illustrates the three possible outcome patterns. If no metric exceeds the noise trajectory ($p > 0.05$ at all layers), no convergence is detected beyond architectural baselines, consistent with Zamir’s hypothesis. We note that absence of signal does not formally prove fragmentation: the metrics may miss forms of convergence outside their scope. If deep-layer metrics remain significant ($p < 0.05$), we have strong evidence that the encoder maps distinct inputs into a partially shared space. However, we cannot determine whether remaining unaligned variance is legitimate modality-specific information or a unification failure (ceiling ambiguity, addressed by RQ2).

Controls. (a) We report VQ-VAE tokeniser reconstruction quality per modality to assess input-quality confounds. (b) We estimate per-modality training exposure frequencies (mono-modal RGB is far more frequent than depth) and discuss whether metric differences correlate with this asymmetry rather than fragmentation.

B. Exploratory Protocols (RQ2)

RQ1 leaves two blindspots: functional compatibility and ceiling ambiguity. Below we sketch candidate protocols that *may* address these gaps. These preliminary ideas are exploratory and will be refined as the project develops; part of our contribution will be to assess whether they yield interpretable results.

1) *Functional Compatibility*: Geometric similarity does not guarantee that shared information is encoded *interchangeably*. Possible approaches include: (a) *Cross-modal k-NN*: retrieve the k nearest RGB neighbours of a depth representation; semantically consistent neighbours would suggest functional

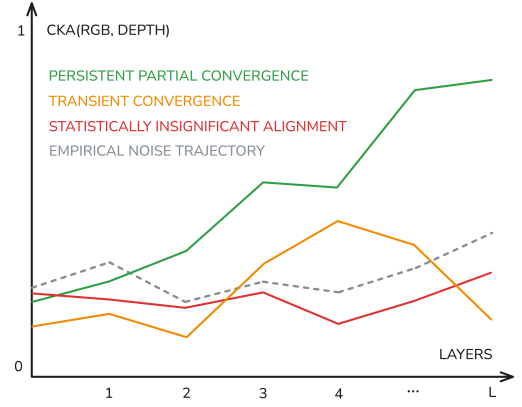


Figure 2. RQ1 expected outcomes: layer-wise CKA between aligned RGB and depth inputs, compared against an empirical noise trajectory (dashed, from mismatched scenes). Green: persistent partial convergence (significant above noise in deep layers). Orange: transient convergence (emerges then collapses). Red: no detectable convergence (indistinguishable from noise).

compatibility without any learned parameters. (b) *Cross-modal decoding*: using 4M’s native any-to-any capability, compare RGB→depth quality against depth→depth; a small gap would suggest functional interchangeability. Whether these tests can isolate encoder unification from decoder effects remains to be verified.

2) *Ceiling Ambiguity*: Significant convergence in RQ1 leaves open whether unaligned variance hides shared information. Possible approaches include: (a) *Residual correlation*: project out the shared subspace ($Z_{RGB}^\perp$) and test whether its distance structure correlates with ground-truth depth distances. (b) *Transmodal triangulation*: measure CKA between the RGB-depth residual and a third modality (normals); non-zero alignment would indicate shared geometry left fragmented. Both approaches assume that orthogonal projection is meaningful in high-dimensional, non-linear spaces, which requires validation.

III. RELATED WORK

Any-to-any models. 4M [1]/4M-21 [7] and UnifiedIO-2 [8] unify modalities but do not assess representational unity. The Platonic Representation Hypothesis [3] predicts convergence across models; we test it *within* one model. **Modality gap.** Liang et al. [9] study gaps between *separate* encoders; we study pathways within a *single shared* encoder. **Representational similarity.** CKA [5] and PWCCA [6] are standard for comparing representations across models. We apply them in a novel setting: the same encoder fed different modalities of the same scene, with null hypothesis testing against mismatched scenes.

Feasibility. All resources are public: 4M-B/L on HuggingFace, Hypersim [10] (synthetic, all modalities), DIODE [11] (real, RGB/depth/normals). RQ1 requires forward passes only (~ 50 – 100 GPU-hours). **Timeline:** W1–2 RQ1 pipeline; W3–4 full analysis + midterm; W5–6 RQ2 explorations; W7 deliverables.

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